3. App Flow — TCC Event Clearance App

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Every screen sequence for students, department SBO officers and the head officer, including the offline steps and failure cases.

Install gate and sign-in

The app does nothing until it runs as an installed app. Desktop screens (the officer dashboard) are exempt.

1. The student opens the link, shared through a QR poster or a message.
2. The app checks two things: is it running installed (standalone mode), and is it inside an in-app browser such as Messenger or Facebook.
3. Inside an in-app browser, the screen says "Open in Safari or Chrome" and offers a copy-link button, because those browsers cannot install anything.
4. On Android Chrome, not installed: a large "Install to continue" button triggers the browser's install prompt.
5. On iPhone Safari, not installed: a picture guide shows Share, then Add to Home Screen, then open the app from the home screen.
6. Once installed, the student continues to Google sign-in.

On iPhone the installed app and Safari keep separate storage, so registration always happens inside the installed app.

Student registration and approval

Registration needs internet once, so plan registration days at school wifi.

1. Sign in with Google.
2. Read and accept the consent screen: what is collected, why, and when it is deleted.
3. Enter name, school ID, department, year level and section. A school ID can be registered only once.
4. Take the live photo. The front camera opens with an oval guide, and there is no gallery option. The device checks for exactly one clear, centered face that is bright and sharp enough, and shows "make sure your face is clearly visible" otherwise.
5. The app creates a device key pair; the public key is sent with the registration.

6. The student sees "Waiting for approval from the [department] SBO".
7. The department SBO opens a pending list for their own department (name, school ID, section, photo) and approves, or rejects with a reason.
8. Approved: the student can download event packs and show a QR. Rejected: the student corrects the details and resubmits.

Student flow on event day

Before the event (needs internet once)

1. Open the app and see the list of published events for the student's department.
2. Tap "Download event pack" to save the card design, details, venues and officer public keys to the phone.
3. A green "ready for offline" badge appears when the download is complete.

At the event (no internet needed)

1. Open the event. If the event has a location check, allow location once; a "finding your location" message appears with a retry button.
2. The QR screen shows the student's name and a QR that refreshes every 30 seconds.
3. The student shows the QR; the SBO scans and accepts it.
4. The student's card shows the stamp after the next sync. To get it immediately, the student taps "Get stamp" and scans the officer's receipt QR.
5. After the event, the student can export the card as an image or PDF.

If something is wrong

- Photo rejected at the scanner: a banner shows the reason and a Retake button, and the QR stays hidden until the new photo is taken.
- Manual check-in by an SBO: the slot appears on the card marked "manual".

SBO flow: create, design and publish an event

1. Tap New event and enter the name, description, school year and the departments it is for.
2. Set the days and the slots per day. Each slot has a label (Time In, Time Out or custom) and a time.

3. Add venues by searching or dropping a pin on the map; the officer's own location does not matter. Set a radius per venue and choose the location check: Off (default), or On with Flag only (default when on) or Block.
4. Choose which officer signs each slot; the signer's name and position print under the signature line.
5. Design the card: pick a preset, or enter a theme and let AI generate artwork options (limited regenerations per event). The app builds the layout and writes all text itself. Edit fields, colors and positions, then preview.
6. Show the draft to the instructor in person. The instructor approves the details and design.
7. Enter "Approved by" (instructor name) and the date.
8. The head officer publishes. Students and scanners can then download the event pack.

If an event changes after publishing, scanners and students see "event updated, please sync" and keep using the old data until they sync.

SBO flow: scanning, manual check-in and check card

Prepare (needs internet)

1. Sign in; the scanner phone receives its own signing key.
2. Draw your signature once on the signature pad (full white screen) and save it.
3. Choose the event and download the roster, filtered to your department, with photos.
4. Wait for the green "ready for offline" state.

Scan (no internet needed)

1. Pick the venue and the slot. A large banner always shows both, for example "Scanning at: Location 2, Day 1 Morning In".
2. Point the camera at the student's QR.
3. The result screen shows the photo, name, department and section large. Green means accepted. Amber means a flag for the SBO to decide (location does not match, student not eligible, already scanned, phone clock off). Red means an invalid or expired QR.
4. A green result is saved as a signed record on the phone after about 1.5 seconds and the scanner moves on, unless the SBO taps Reject first; an amber result waits for an explicit Accept or Reject. Reject asks for a reason (photo is not a face, does not

match) and forces the student to retake the photo.
5. A counter shows "X scans waiting to sync". Records upload when signal returns; "Export backup" saves them as a file if sync fails.

Manual check-in

Search the roster by school ID or name, tap Manual check-in, and optionally note that a physical stamp was given on the printed card. The record is logged as manual.

Check card

Scan the QR on a student's card, or search by school ID. The screen lists every day and slot as signed (time and signer) or missing, in red, using the stored records and not the card's appearance.

Device reset and edge cases

Device reset (new or lost phone)

1. The student installs the app on the new phone and signs in with the same Google account.
2. The student takes a fresh live photo; the app sends a reset request (needs internet once).
3. The department SBO compares the new photo with the original and approves or rejects; in doubtful cases the SBO asks to see the school ID in person and taps Reset device.
4. Approval revokes the old key and binds the new phone. Until then, the student uses manual check-in.

Edge cases

Situation	What happens
Dead phone, broken camera, no smartphone	Manual check-in plus a handwritten signature on the printed card
Storage cleared on the phone	The device key is lost, so the student requests a device reset
Wrong phone clock	The scanner accepts a small difference and otherwise shows "this phone's clock is off"

Situation	What happens
Student scanned at two gates while offline	On sync the earliest scan is kept and the later one is flagged as a duplicate
Student not in the event's departments	Amber flag; the SBO decides
Location permission refused (check on)	The QR stays hidden; the student goes to an SBO for manual check-in
Sync fails	The officer exports unsynced scans as a file
Student disputes a missing signature	The SBO checks the attendance record on the dashboard, then reissues the stamp or voids the record; the action is logged